



PLL Harmonic Oscillator

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The PLL Harmonic Oscillator (PLLHO) is an audio VCO that produces ramp outputs in harmonic relationships. The v.7 implementation produces the first 8 harmonics, although the concept can be extended to any number. The exact tuning to harmonic frequencies is achieved with two different mechanisms which operate in conjunction - control current coupling and sampled-ramp PLL.

1. Control current coupling

The oscillator has 8 simple analog OTA-based ramp cores that share the same control current. The implementation incorporates a standard tempco/BJT exponential converter (see Board A - Sheet 2 schematic) with 8 output transistors in parallel. Only the transistor providing bias current to the fundamental core is matched and thermally coupled to the compensation transistor and tempco resistor, others are not, but the copies of the current that they provide are close enough.

The integration capacitors are chosen to produce frequencies that approximate harmonics. Also, the cores can be fine-tuned by varying the OTA input voltages. This is more-or-less the same as using 8 separate VCOs, tuning them as harmonics and controlling all 8 with the same CV. It also exhibits the same issues - the frequency relations drift due to various factors and do not persist over a large frequency range. However, the native frequencies are always near the target harmonics - 15% error or less. This is where SRPLL comes in.

2. Sampled-ramp PLL

Classic PLL (like 4046) takes a pulse waveform as control input, and the tracking VCO also outputs pulses. The common methods of phase error detection are digital - XOR gate, edge-triggered flip-flop and similar. The error detectors compare phase offsets between control and tracking VCO signals and output pulses of variable duration. These pulses are integrated in a low-pass filter to produce a more-or-less continuous error voltage which is used to correct the tracking VCO frequency.

SRPLL control input is a zero-centered ramp with fixed amplitude. The tracking VCO output is also ramp, but it could be any other shape as long as the core produces a reset pulse. The error voltage output is not continuous, it's a series of discrete values.

The advantage of this system is very fast locking, typically in 3 control input cycles or less, with the first cycle approximation very close to F_l . Longer settling times of classic PLL make it unusable for lower-frequency audio.

The disadvantage is a small locking frequency range, about 1.5 octaves maximum, but this is compensated with control current coupling (1.), which always shifts the native frequency into the locking window.

The tracking oscillator ramp resets are driving a counter with decoded outputs, e.g. CD74HC4017. The length of the cycle determines the desired locked frequency, F_l :

$$F_l = N \cdot F_c$$

where N is the number of steps in the cycle and F_c is the control input frequency.

Once per cycle, on one of the tracking oscillator resets, the control ramp voltage is sampled and held. This is error voltage, V_e , and it's directly proportional to phase error. V_e is fed back to the tracking oscillator frequency control, raising or lowering it. The loop locks when V_e settles to a constant value and the tracking frequency equals F_l .

Here is an example of how SRPLL produces the 4th harmonic. The control input is H1 with frequency F_c and the locked H4 frequency is $F_l = F_c \cdot 4$. Both oscillators start in sync. The control ramp will be sampled on step #2 of the counter cycle.

First consider a tracking VCO with a native frequency that is exactly the same as the desired locked frequency - $F_n = F_l$ (Figure 1.)

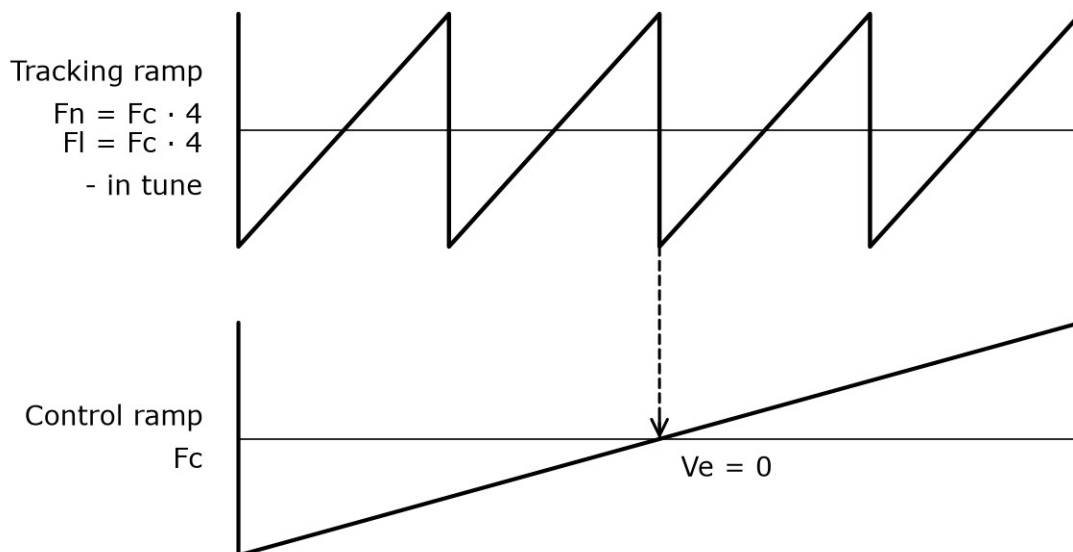


Figure 1.

The sampled error voltage $V_e = 0$, so the tracking oscillator frequency is not changed – it's already locked.

Figure 2. shows the case with native tracking frequency $F_n = F_c \cdot 3$ (too low). This is the ramp above the control ramp. The sampled error voltage on reset #2 is a positive value. This V_e is passed to the tracking VCO's frequency control. Presuming an ideal correction gain, V_e will raise the tracking frequency to $F_c \cdot 4$, which is F_l . Subsequent sampling of the control ramp will always yield the same value - frequency is locked. This is the ramp below the control ramp.

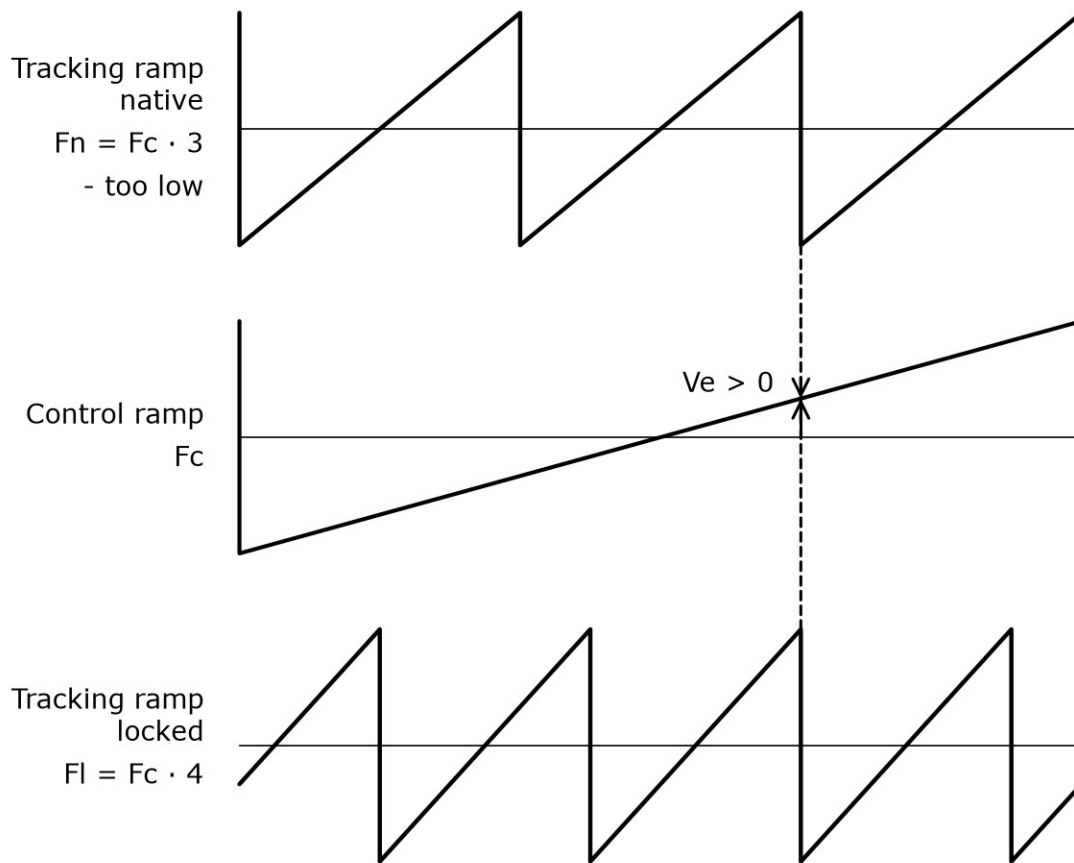


Figure 2.

Note that the phase offset between the two ramps is changed.

Figure 3. shows the same thing with native tracking frequency $F_n = F_c \cdot 7$ (too high). The sampled error voltage on reset #2 is now a negative value. Note that the phase is now offset in the opposite direction.

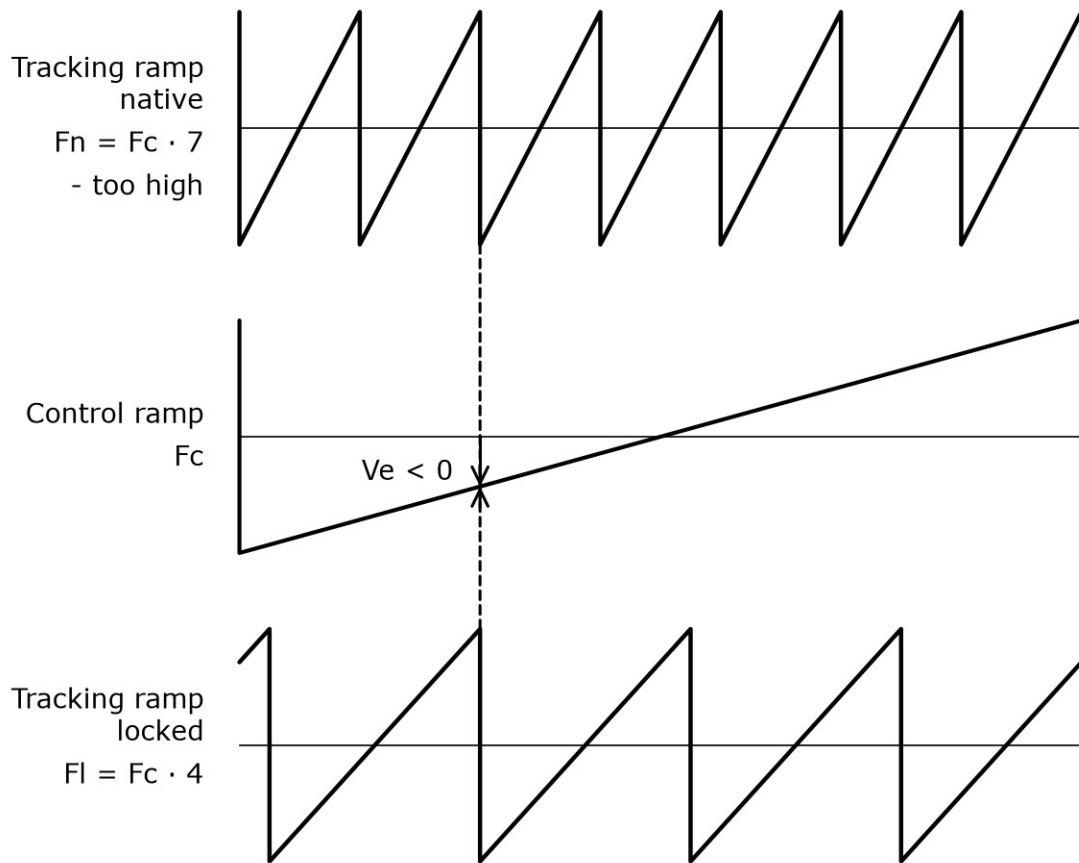
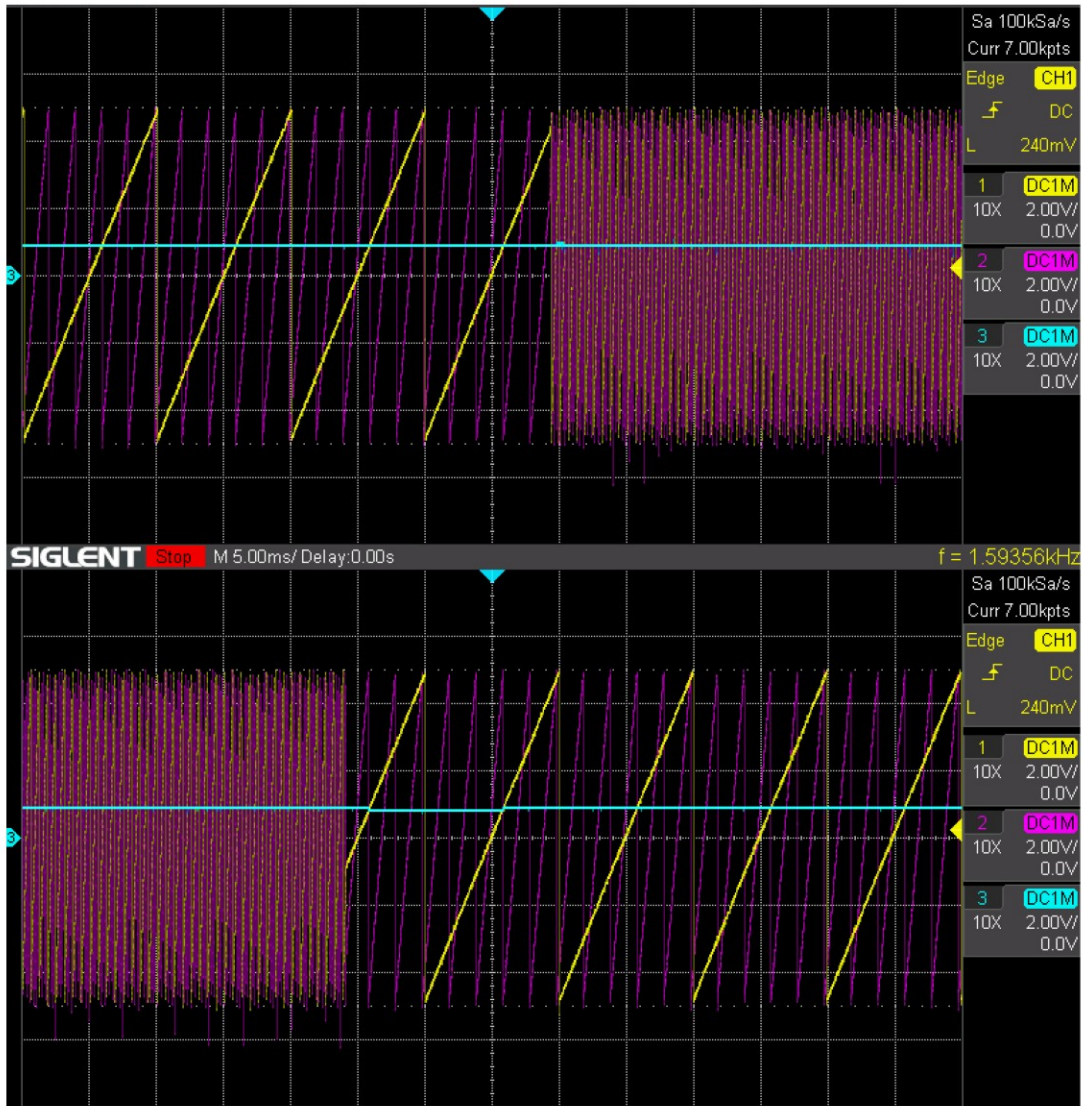


Figure 3.

This example is simplified and assumes ideal conditions but should paint the picture. The correction actually takes place immediately every time V_e is sampled - the graphs gloss over that. The correction gain is not ideal, there's always some overshoot/undershoot, so lock on first sample is rare. The implemented circuit, with control current coupling, settles in 3 fundamental cycles or less.

Note that in the implementation only H5/H6/H7/H8 harmonics are locked using PLL (see Board B - Sheet 2 and Board C - Sheet 2 schematics). H2/H3/H4 are slaves, getting the same control bias, but the resets are divided down from H6 and H8 cores (see Board D - Sheet 2 schematic). The slave OTAs get the same input voltages as the master cores OTAs (scaled).

Here are scope captures made with the v.7 implementation, showing the 5th harmonic settling period. The yellow trace is the fundamental waveform, pink is the harmonic and blue is the sampled error voltage.



Steps 100Hz - 1.6kHz - 100Hz

Figure 4.

H1 step in the upper capture is 100Hz to 1.6kHz (4 octaves), 1.6kHz to 100Hz in the lower one. Ve in the lower capture has locked on the second fundamental cycle, so settling time is less than 20ms at 100Hz fundamental.

3. Phase shift and modulation

The example in (2.) demonstrated how the phase offset between control and tracking ramps depends on the native frequency error. A flat native ramp will shift phase to the right, a sharp native ramp will shift left. This also means that we can change the phase offset between H1 and other harmonics intentionally, by detuning tracking oscillators. The actual amount of phase shift that can be achieved depends on a number of factors, and may be different for different harmonics, but usually goes up to $\pm 180^\circ$ relative to fundamental.

In the v.7 PLLHO implementation, phase offset CV control is applied in two modes: Mode A shifts H2 to H8 in the same direction, Mode B shifts two groups of harmonics in opposite directions. Since this involves detuning the oscillators, phase offset CV is conditioned so PLL will not break for most of the input range. But it's not clamped, you can break PLL if you feel like it.

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